Homework 08

The former kernel-tuning question has moved to Homework 09.

Solutions for Questions 1 and 2 will be available after Monday, October 19, 2026. No solution can be provided for Question 3.

For Questions 1 and 2, use either R or Python.

Question 1: Boundary and margin

Use `margin-geometry-20.csv` from the combined ZIP, which contains 20 observations with predictors x_1, x_2 and class labels $y \in \{-1, 1\}$. Consider the score

$$f(x) = 0.8x_1 + 0.6x_2.$$

Since $\|(0.8, 0.6)\|_2 = 1$, the lines $f(x) = \pm 1$ are each a distance 1 from the decision boundary $f(x) = 0$.

a. Calculate the distance from every observation to the decision boundary and its functional margin $yf(x)$. In a table, report these values and whether each observation is inside, on, or beyond the margin for its class. Compare $yf(x)$ with 1 to make this distinction.

b. Plot the observations, the boundary $f(x) = 0$, and the margin lines $f(x) = \pm 1$. Distinguish the two classes and identify the observations inside the margin in your plot.

c. Report how many observations are inside, on, and beyond the margin. Which observations are misclassified by this decision rule?

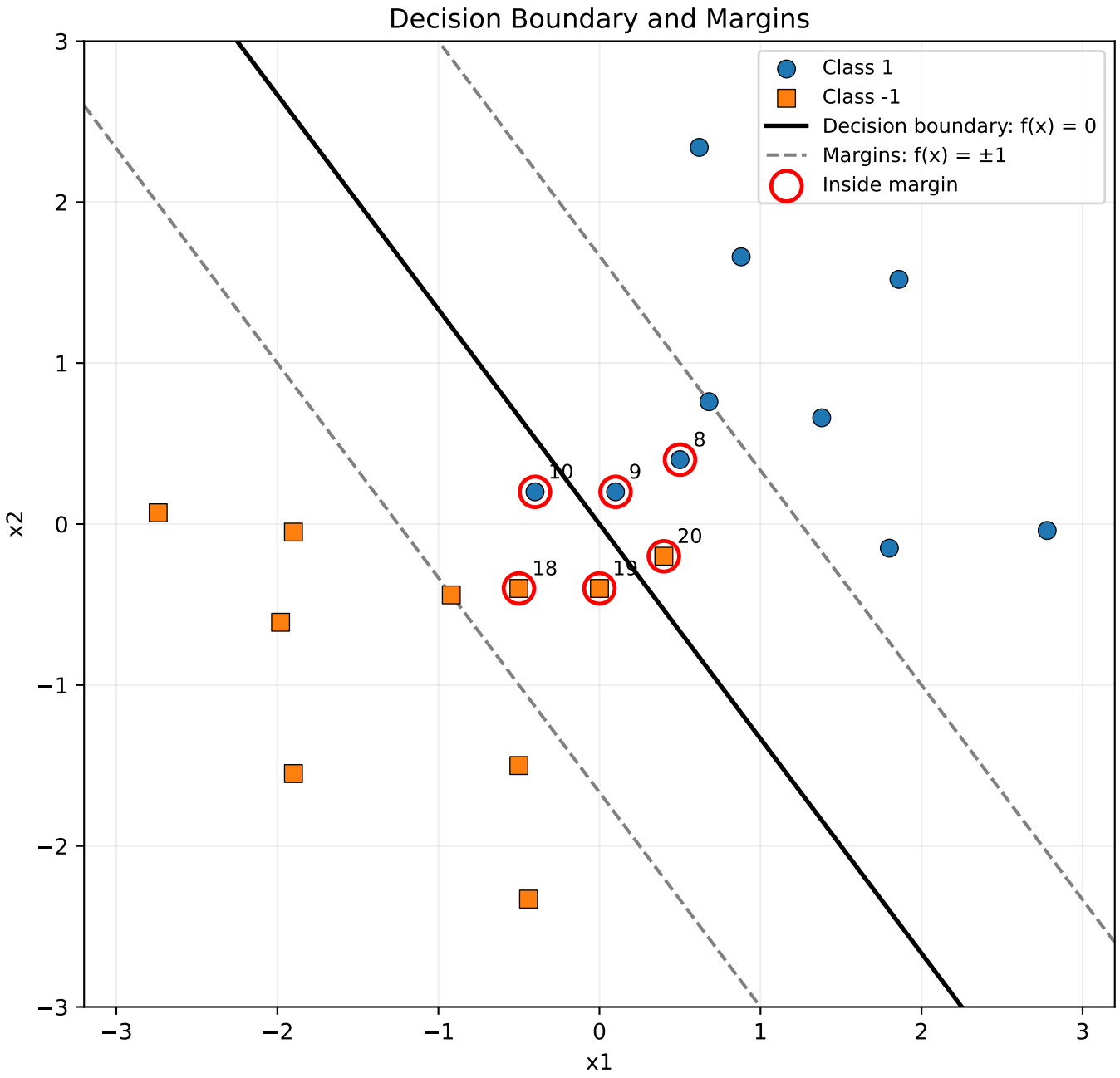
Use the observation numbers supplied in the data. Set the x_1 axis to $[-3.2, 3.2]$, set the x_2 axis to $[-3, 3]$, and use an equal aspect ratio. Do not fit a new classifier for this question.

Answer

Part a

```
Part (a): Results
 observation    x1    x2  y  score  distance  functional_margin margin_status  misclassified
           1  0.68  0.76  1   1.00      1.00               1.00            On          False
           2  1.80 -0.15  1   1.35      1.35               1.35        Beyond          False
           3  1.38  0.66  1   1.50      1.50               1.50        Beyond          False
           4  0.88  1.66  1   1.70      1.70               1.70        Beyond          False
           5  0.62  2.34  1   1.90      1.90               1.90        Beyond          False
           6  2.78 -0.04  1   2.20      2.20               2.20        Beyond          False
           7  1.86  1.52  1   2.40      2.40               2.40        Beyond          False
           8  0.50  0.40  1   0.64      0.64               0.64        Inside          False
           9  0.10  0.20  1   0.20      0.20               0.20        Inside          False
          10 -0.40  0.20  1  -0.20      0.20              -0.20        Inside           True
          11 -0.92 -0.44 -1  -1.00      1.00               1.00            On          False
          12 -0.50 -1.50 -1  -1.30      1.30               1.30        Beyond          False
          13 -1.90 -0.05 -1  -1.55      1.55               1.55        Beyond          False
          14 -0.44 -2.33 -1  -1.75      1.75               1.75        Beyond          False
          15 -1.98 -0.61 -1  -1.95      1.95               1.95        Beyond          False
          16 -2.74  0.07 -1  -2.15      2.15               2.15        Beyond          False
          17 -1.90 -1.55 -1  -2.45      2.45               2.45        Beyond          False
          18 -0.50 -0.40 -1  -0.64      0.64               0.64        Inside          False
          19  0.00 -0.40 -1  -0.24      0.24               0.24        Inside          False
          20  0.40 -0.20 -1   0.20      0.20              -0.20        Inside           True
```

Part b



Part c

```
Part (c): Margin counts
margin_status
Inside     6
On         2
Beyond    12
Name: count, dtype: int64

Misclassified observations:
    observation  functional_margin
9            10               -0.2
19           20               -0.2

Misclassification rate:
10.00%
```

Question 2: Margin in four dimensions

Use `margin-geometry-4d.csv` from the combined ZIP, which contains 50 observations with four predictors and class labels $y \in \{-1, 1\}$. Consider the linear score

$$f(\mathbf{x}) = \beta_0 + \mathbf{x}^\mathsf{T}\beta, \qquad \beta_0 = -1, \quad \beta = (1, 2, -2, 4)^\mathsf{T}.$$

a. Project the observations onto the direction of β and plot their positions on one axis. Distinguish the two classes and mark the decision boundary. Use the plot to calculate the geometric margin M, the distance from the boundary to the closest observation.

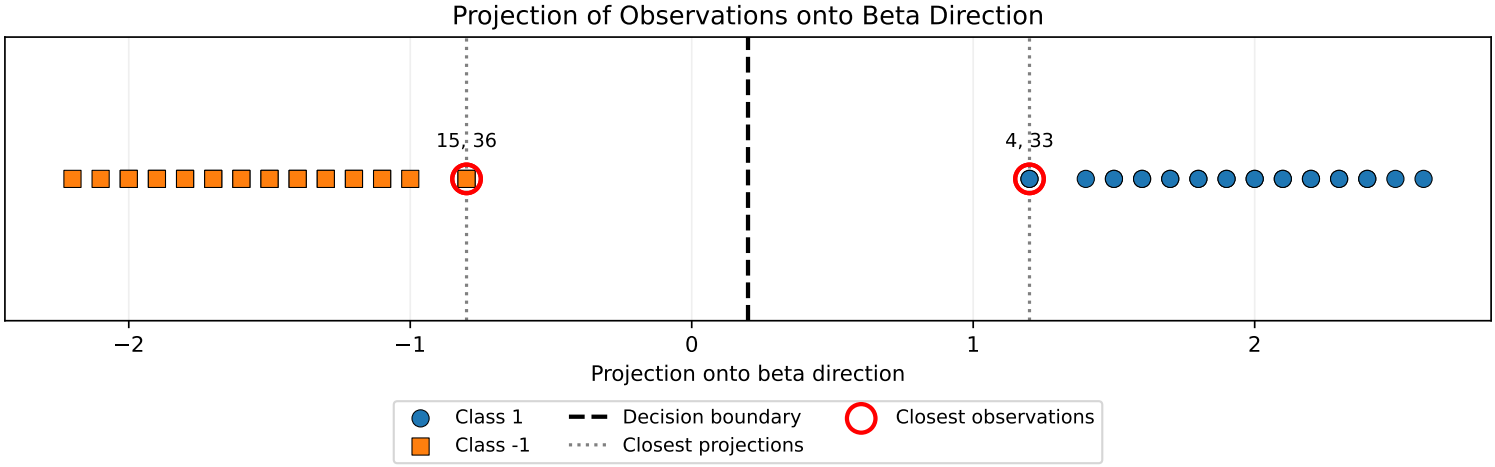
b. Express the same decision boundary using the SVM normalization in which the smallest functional margin $y_i f(\mathbf{x}_i)$ is set to 1. What are the corresponding values of β and β_0?

Answer

Part a

```
Norm of beta: 5.0
Projected boundary: 0.2
Geometric margin M: 1.0

Closest observations:
 observation    x1    x2    x3    x4  y  projection  score  functional_margin  distance
           4 -1.26 -0.27 -1.98  0.96  1         1.2    5.0                5.0       1.0
          15 -6.16  1.18  0.32  0.11 -1        -0.8   -5.0                5.0       1.0
          33  7.74  1.23  1.02 -0.54  1         1.2    5.0                5.0       1.0
          36  2.84  0.43 -0.43 -2.14 -1        -0.8   -5.0                5.0       1.0
```



Part b

```
Part (b): SVM normalization
Original minimum functional margin: 5.0
Normalized beta: [ 0.2  0.4 -0.4  0.8]
Normalized beta0: -0.2
New minimum functional margin: 0.9999999999999998
Geometric margin after normalization: 1.0
```

Question 3: Peer review of a Homework 7 skill

The instructor's GitHub account, **rqzhu-aide**, will open a **GitHub Issue** in the repository containing your Homework 7 submission. It will link to **one** assigned student's repository.

1. **Use the skill.** Find the complete skill definition in the assigned student's Homework 7 report. Copy it without changes into a local `SKILL.md` file in your own folder. You do not need to install it. Give your agent the path to the file and explicitly ask it to follow that skill while analyzing the `SAheart` data, using the response and predictor coding from Homework 7. Provide the necessary data information, but no additional preferences about methods, figures, model selection, or report style. Let the skill direct the analysis.

   Monitor the time, your token use, and your weekly quota while the agent works. If the run is taking too long or using your allowance too quickly, you may send a **steer message** asking the agent to finish immediately, or stop the run and summarize what you observed. A steer message redirects an agent while it is working; learn how to send one in your AI tool before starting.

2. **Write a peer review.** Read the student's submitted report and what the skill produced in your run. Give a brief overall assessment, followed by **three major items the author could improve**. You may comment on any aspect of the skill or report. For each item, explain what you observed, why it matters, and what you recommend changing. If you stopped the run early, explain where and why. Write the review yourself, without AI assistance.

3. **Share your review.** Open a GitHub Issue in the assigned student's Homework 7 repository containing your review. Copy the same review into your Homework 8 submission report and include a link to the GitHub Issue.

Answer

Assigned repository: hethp/stat432-fall2026 (Homework 7), assigned in Issue #3.

Step 1: Using the skill

The complete `SKILL.md` (4006 bytes) was copied without changes from the Homework 7 report into a local file. The agent (Claude Code, Claude Opus 5.5) was asked to follow it on `saheart.csv` with this information only: response `chd`; predictors the other nine variables, with `famhist` coded 1 = Absent, 2 = Present; seed 2026 (the skill requires a seed as an input). The run took about 1.5 minutes (19:36:44 to 19:38:16) and was not stopped early. Usage after the run (Claude Pro plan): 3% of the 5-hour session limit and 13% of the weekly limit. These figures also include the earlier Questions 1–2 work in the same session.

Result produced by the skill: stratified 346/116 train/test split; 5-fold CV mean misclassification rates were LDA 0.2860, QDA 0.3150, PCA(k=8)+LDA 0.2717, PCA(k=6)+QDA 0.2861, Blend($\alpha = 1.0$) 0.2860. The selected model, PCA(8)+LDA, had a test misclassification rate of 0.2414 (confusion matrix: true 0 $\rightarrow$ 65/11, true 1 $\rightarrow$ 17/23).

Step 2: Peer review

Overall Assessment

I found the skill well organized and easy to follow. I particularly appreciated the detailed analysis workflow, including the candidate models, parameter grids, and cross-validation procedure. The instruction to perform standardization and PCA within each CV fold is especially important because it helps prevent data leakage. I also liked the emphasis on a concise one-page report with both visual and numerical comparisons.

Three Major Suggestions for Improvement

1. Clarify how optional models should be included

Observation: The skill requires ordinary LDA and QDA but makes PCA-based models and probability blending optional. It does not specify when the agent should include or exclude these extensions.

Why it matters: Different agents may evaluate different sets of models, making results harder to compare even when analyzing the same dataset.

Recommendation: Add a clearer rule for choosing optional models, such as evaluating all specified candidates when computationally feasible or documenting why certain models were excluded.

2. Improve reproducibility across runs

Observation: Your original analysis selected PCA(6)+LDA with a CV error of 0.2659 using seed 43251. My run selected PCA(8)+LDA with a CV error of 0.2717 using seed 2026. I appreciated that both runs followed the same general approach and selected PCA+LDA.

Why it matters: Different seeds can naturally produce different results, but the skill does not explicitly require recording the CV shuffle settings or how the folds are generated. These details are useful for reproducing an analysis.

Recommendation: Specify the CV randomization procedure and require the report or accompanying code to record the split and fold-generation settings.

3. Strengthen the interpretation of model comparisons

Observation: The skill selects the model with the lowest mean CV error and asks for only 2–3 sentences of findings and limitations. It does not explicitly require discussing how close the leading candidates are.

Why it matters: Small differences in CV error may not indicate meaningful differences in predictive performance. Selecting the numerical minimum alone can make one model appear more clearly superior than the evidence supports.

Recommendation: Require the findings section to compare the top models and acknowledge when their CV errors are very similar relative to the variation across folds.

Final Comments

Overall, I think this is a thoughtfully designed and practical skill. Its clear workflow, predefined tuning grids, leakage prevention, and compact reporting format are strong features. My suggestions mainly aim to improve consistency across runs, reproducibility, and interpretation rather than change the overall approach.

Step 3: GitHub Issue

The review above was posted as a GitHub Issue in the assigned repository: hethp/stat432-fall2026#5.